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(Not) Thinking About the Future: Financial Information and Maternal Labor Supply

Contents

1 Big picture

- **Question.** Does objective information about the long-run financial costs of reduced labor supply change mothers' work plans and actual working hours?
- **Core setting.** Switzerland has high maternal employment but unusually high rates of low part-time work. The main experiment studies part-time mothers employed as public school teachers in a large German-speaking region.
- **Main economic friction.** Mothers may not fully internalize the delayed financial consequences of low hours. The key costs are not only current earnings but also lifetime earnings, returns to experience, occupational pension accumulation, and financial independence after adverse events.
- **Attention friction.** Long-run financial factors are not top of mind. In the descriptive survey, most mothers emphasize children, household organization, job fit, family preferences, or short-run budget constraints rather than pensions or lifetime earnings.
- **Belief friction.** A substantial share of women holds overly optimistic beliefs about pension receipt or wage growth under low part-time work. The paper calls these women *cost underestimators*.
- **Intervention.** Treated mothers receive a short informational video and access to personalized projections from the Future Calculator. The treatment quantifies how reduced hours affect lifetime income, occupational pensions, and financial well-being.
- **Design.** The experiment uses two-stage randomization. One-third of schools are pure controls. In the remaining schools, teachers are individually randomized to treatment or within-school control, allowing the authors to estimate both direct effects and possible social spillovers.
- **Main first stage.** The treatment improves financial understanding and increases demand for financial planning tools, especially among cost underestimators.
- **Main labor supply result.** Cost underestimators increase actual employment by 3.87 percentage points one year later, about 7% over the pure-control mean.
- **Long-run implication.** If the employment increase is sustained, it would reduce the teacher gender gap in lifetime income and pension receipt by about 18%.
- **Asymmetry.** Women who overestimate the costs of part-time work do not symmetrically reduce hours after learning they are less financially harmed than expected. This asymmetry is consistent with the intervention being more informative for the group that underestimated costs.
- **Takeaway.** Information frictions can be an economically meaningful labor supply friction. Policies that reduce constraints such as childcare may be less effective if families do not perceive the long-run returns to increasing hours.

2 Data and measurement

2.1 Swiss maternal labor supply and pension context

Switzerland combines high maternal employment with persistent low part-time work. Mothers frequently remain attached to the labor market, but they often work at low fractions of full-time equivalent employment. This matters because reduced hours lower current income and also reduce long-run pension accumulation and returns to experience.

The pension system is especially important. The paper focuses on the second-pillar occupational pension system, where contributions depend on earnings above a threshold and where accumulated funds are individual and portable across employers. Low hours can sharply reduce occupational pension receipt because contributions scale with earnings and because part-time work can keep workers closer to minimum thresholds.

Teachers are a useful proof-of-concept population because salary scales are deterministic, promotion uncertainty is limited, and annual hours can be adjusted in small increments. These features allow the researchers to construct credible personalized projections and then observe whether mothers act on them during the yearly planning cycle.

Interpretation: The setting is intentionally chosen to make the information accurate and actionable. This strengthens internal validity, but also means that the results should be extrapolated carefully to jobs with less predictable wage growth, less flexible scheduling, or more nonlinear career ladders.

2.2 Descriptive survey and top-of-mind factors

The descriptive survey targets mothers in German-speaking Switzerland and asks open-ended questions about what they considered when choosing labor supply after childbirth. It also uses a vignette about a part-time mother considering an increase in employment level.

The key descriptive measure is whether long-term financial factors are mentioned without prompting. This is different from whether mothers care about money at all. The paper's point is that long-run financial consequences are often not salient in the decision environment, even if immediate financial constraints are considered.

Interpretation: The open-ended design is important because it captures salience rather than prompted knowledge. A mother may understand that pensions matter when asked directly, but still not spontaneously include pensions in her actual labor supply choice set.

2.3 Vignette, Future Calculator, and cost underestimation

The vignette elicits beliefs about current salary, future salary, pension receipt, and the financial value of moving from low part-time work to a higher employment level. The authors adjust the vignette values by education group and use the Future Calculator to produce benchmark projections.

The paper measures two main kinds of misperception:

- **Pension underestimation of costs:** respondents overestimate pension receipt under low part-time work.
- **Wage-growth underestimation of costs:** respondents underestimate the wage-growth penalty from remaining at low hours.

Cost underestimators are mothers whose baseline beliefs make low part-time work look financially less costly than the calculator indicates. The classification is made before treatment and is therefore not itself caused by treatment exposure.

Interpretation: This predetermined heterogeneity is central to the paper. If the treatment works by correcting misbeliefs, the largest effects should appear for mothers whose prior beliefs make the information novel and corrective. This is exactly the group that responds most strongly.

2.4 RCT sample, treatment arms, and outcomes

The main RCT sample contains 2,359 non-full-time mothers across 479 schools. Teachers receive either the financial-information treatment or placebo content. The outcome set is deliberately broad:

- understanding and ranking of financial factors;
- demand for financial tools and financial consultation;

- short-run employment plans for the next academic year;
- medium- and long-run employment plans at three, five, and ten years;
- administrative employment one year after the intervention;
- mechanism outcomes such as emotions, discussions, actions, partner plans, and fertility plans.

Interpretation: The combination of survey and administrative outcomes is a major strength. Survey outcomes reveal belief updating and planned reoptimization, while employer administrative data show whether those plans translate into actual contracted hours.

2.5 Administrative employer data

The survey is linked to Department of Education personnel data for 2020–2023. The main realized outcome is employment level as a share of full-time equivalent employment in the year after treatment.

The administrative data cover the primary employer for most teachers, and the paper reports a high match rate between survey respondents and administrative records. The data are still not perfect: they do not fully observe non-teaching jobs outside the Department of Education. The authors argue this limitation is likely modest because most female teachers with children work exclusively as public school teachers and outside jobs are relatively small for those who have them.

Interpretation: Administrative data are crucial because they remove concerns that the main result is only an intention or survey-demand effect. The behavioral outcome is a realized change in contracted hours.

3 Models and empirical specifications

3.1 Baseline treatment and spillover regression

For every primary outcome, the paper estimates a specification of the form

$$Y_{is} = \beta_0 + \beta_1 Treat_{is} + \beta_2 Spillover_{is} + \beta_3 X_{is} + \beta_4 X_s + \gamma_f + \varepsilon_{is}.$$

Here i indexes the teacher, s indexes the school, $Treat_{is}$ indicates direct treatment, and $Spillover_{is}$ indicates an untreated teacher in a treatment school. X_{is} contains individual baseline controls, X_s contains school controls, and γ_f are stratification fixed effects.

Interpretation: β_1 measures the direct treatment effect relative to teachers in pure-control schools. β_2 measures whether untreated teachers in treated schools are indirectly affected through colleague discussion or other social learning.

3.2 Heterogeneity by cost underestimation

To test whether learning drives the results, the paper estimates interacted specifications of the form

$$Y_{is} = \alpha + \beta_U Treat_{is} \times Under_i + \beta_O Treat_{is} \times Over_i + \gamma_U Spillover_{is} \times Under_i + \gamma_O Spillover_{is} \times Over_i + \kappa Under_i + X'_{is}\pi + X'_s\rho + \delta_f +$$

$Under_i$ indicates that the mother underestimated the costs of part-time work at baseline; $Over_i$ or the comparison group captures those who did not underestimate costs.

Interpretation: This specification asks whether treatment effects are concentrated among mothers whose priors make the information corrective. A strong β_U and weak β_O supports belief updating rather than generic encouragement to work more.

3.3 Administrative labor supply outcome

The main realized labor supply outcome is

$$\Delta Employment_i = Employment_{i,2023} - Employment_{i,2022}.$$

Employment is measured as percent of full-time equivalent employment. The interpretation is literal: a positive coefficient means that the contractually recorded employment level rises after the intervention.

Interpretation: The outcome is tied to the institutional timing of the intervention. Teachers decide yearly hours during the planning period, so the administrative outcome is a direct test of whether information changes a real labor supply decision.

3.4 Financial-information index and planning-tool demand

The paper aggregates financial outcomes into a prespecified financial index. The components include whether the teacher correctly ranks long-term financial factors relative to childcare costs and whether she demands tools such as personalized financial information or consultation.

Interpretation: The index separates two related channels. First, the intervention can change understanding. Second, it can raise demand for follow-up planning tools. The stronger response among cost underestimators in tool demand indicates that corrected beliefs make additional information more valuable.

3.5 Nonparametric heterogeneity

The paper also estimates nonparametric treatment effects by baseline pension estimate. Respondents are binned by their pension estimate, and treatment effects are estimated locally across the distribution.

Interpretation: This is a visual mechanism test. If the effect is truly driven by mistaken beliefs, it should rise smoothly among women whose baseline estimates are more optimistic relative to the true projected pension value, rather than appearing only because of an arbitrary binary cutoff.

3.6 Inference and controls

The paper uses strata fixed effects and post-double-selection LASSO to select controls from baseline individual variables and school-level variables. Standard errors are clustered at the school level, and sharpened q-values are reported for multiple-hypothesis adjustment in the tables.

Interpretation: These choices address three practical issues: randomization strata, many possible baseline controls, and multiple related outcomes. They do not substitute for random assignment, but they improve precision and discipline the interpretation of statistical significance.

4 Identification and empirical design

4.1 Core randomization logic

Treatment assignment is randomized, so baseline potential outcomes should be independent of treatment status within strata. The two-stage design creates a pure-control group plus a within-treatment-school control group. This design is necessary because teachers within the same school may discuss the intervention.

Interpretation: Pure-control schools provide an uncontaminated counterfactual. The within-school spillover group is not merely a control group; it is also an object of interest because it measures whether treatment information diffuses through social networks.

4.2 Why two-stage randomization matters

A standard individual-level RCT within schools could underestimate the treatment effect if untreated colleagues learn about the video from treated colleagues. Conversely, a school-level design alone would make it harder to separate direct effects from school-level shocks. The two-stage design solves both problems: it estimates direct effects and spillovers while preserving a clean hold-out comparison.

Interpretation: This is a particularly appropriate design because the mechanism includes conversations with partners, colleagues, and social circles. If information is socially transmitted, contamination is not a nuisance only; it is part of the policy-relevant mechanism.

4.3 Timing and actionability

The intervention is implemented at the start of the yearly employment planning period, when teachers communicate desired employment levels to school principals. This timing makes the information actionable. If the same information arrived after the employment planning deadline, it might change beliefs without changing behavior for another year.

Interpretation: The timing strengthens causal interpretation of the administrative employment effect. The treatment arrives precisely when teachers can change the relevant choice margin.

4.4 Predetermined heterogeneity and mechanism test

Cost underestimation is measured before treatment exposure. This avoids post-treatment sorting into mechanism groups. The finding that effects are concentrated among cost underestimators is central: it matches the predicted response of a belief-updating model.

Interpretation: A generic salience or persuasion story would likely affect a broader group of mothers. The narrow concentration among the women whose beliefs were too optimistic supports the interpretation that the treatment corrects information gaps.

4.5 Survey-to-administrative evidence chain

The paper builds an unusually complete chain of evidence:

- descriptive survey evidence shows long-run factors are not top of mind;
- baseline elicitation identifies misperceptions about pension and wage-growth costs;
- random treatment changes financial understanding and demand for tools;
- cost underestimators revise short- and long-run employment plans upward;
- administrative data show actual hours increase for the same group one year later;
- mechanism outcomes show discussion, emotional reaction, and follow-up actions.

Interpretation: The paper is compelling not because of one coefficient alone, but because each link of the behavioral pathway points in the same direction.

4.6 Administrative outcome validity

The administrative employment measure avoids self-report bias and survey experimenter-demand concerns. Because the outcome is recorded by the employer and measured one year after treatment, it is less vulnerable to immediate response bias.

Interpretation: This is especially important because the treatment is informational and could otherwise induce treated respondents to state higher labor supply plans without actually changing hours.

4.7 Spillovers and social learning

The paper treats spillovers as both a design issue and a substantive result. Treated teachers are more likely to discuss the intervention with colleagues, and there is suggestive learning among cost underestimators in the spillover group.

Interpretation: Spillovers imply that information interventions may have multiplier effects if delivered through employers, pension funds, or social networks. At the same time, spillovers also mean that pure-control schools are essential for identifying direct effects cleanly.

4.8 Robustness and external validity

The paper reports robustness checks for experimenter-demand effects, sample and specification choices, timing, and engagement. It also provides suggestive evidence that incremental hour increases are feasible in broader occupations and that similar short-term results appear in a small sample of pregnant women outside teaching.

Interpretation: The robustness evidence supports the internal validity of the main RCT. External validity remains more cautious: teachers have predictable salaries, portable pensions, and flexible annual hours, while other occupations may feature nonlinear promotion paths or less flexible scheduling.

4.9 Remaining limitations

Information does not remove childcare constraints, household bargaining frictions, social norms, or employer constraints. The teacher setting is deliberately favorable for actionability, which helps prove the information mechanism but may overstate the ease of response in less flexible jobs. The paper also focuses on one-year realized outcomes; the longer-run employment and pension impacts are inferred from plans and projections rather than observed over decades.

Interpretation: The study is best read as proof of concept that information frictions matter, not as evidence that information alone can eliminate child penalties or gender gaps.

5 Tables and figures

All visuals appear only in this section. Each image is directly cropped from the paper or the source notes bundle.

5.1 Figure I: "Top of Mind": Factors Considered in Labor Supply Decision After Childbirth

Read:

- The figure reports categories from open-ended responses about what mothers considered when choosing post-childbirth labor supply.
- Child-related factors, household organization, job characteristics, and mother well-being dominate the responses.
- Long-term financial factors such as pensions, lifetime earnings, career progression, or financial independence are mentioned by only a small minority of mothers.
- The key empirical fact is not that mothers ignore all financial matters. Rather, the financial considerations that spontaneously appear tend to be immediate rather than long horizon.

Interpretation: The figure establishes the attention problem that motivates the experiment. If mothers do not have delayed financial consequences in the active decision set, even large long-run costs may not affect hours choices. This is a behavioral friction distinct from preferences for childcare, household specialization, or leisure.

Mechanism role: The figure supports the first step of the paper's mechanism: long-term information is either absent from attention or difficult to compute. The intervention is designed to make that hidden trade-off explicit.

Economic message: Labor supply choices may be distorted not because mothers do not care about future income, but because the future costs of low hours are opaque at the moment of choice.

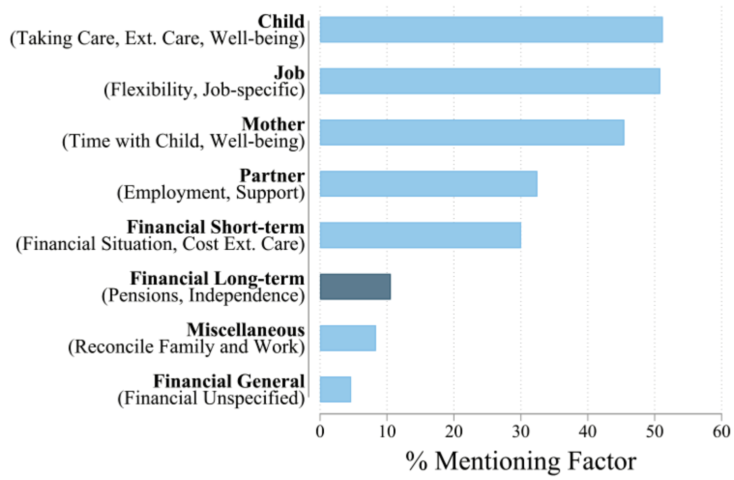


FIGURE I

"Top of Mind": Factors Considered in Labor Supply Decision After Childbirth

This figure shows the percentage of women who mention a given topic when asked which factors they considered for their labor supply decision after the birth of their first child. Coding and validation are documented in Online Appendix B.5. Data: Female sample, descriptive survey. $N = 539$.

Figure I: "Top of Mind" factors considered in labor supply decisions

5.2 Table I: Treatment Impact on Financial Outcomes

Read:

- The table reports treatment and spillover effects on a financial index, the ranking exercise, demand for financial tools, sign-up for consultation, retention of the information in follow-up, and open-ended mention of long-term financial factors.
- In the full sample, the treatment improves financial understanding, especially the ability to rank long-run salary and pension effects above childcare costs.
- Among cost underestimators, the treatment strongly increases demand for financial tools; the paper reports an increase of about one-third of a standard deviation for the tools index.
- In the follow-up survey, treated women remain more likely to apply the treatment information correctly to a similar vignette.
- Spillover effects are generally weaker and noisier, but there is suggestive learning among cost underestimators in treated schools.

Interpretation: Table I is the first-stage table. It verifies that the treatment actually changes the informational environment before the paper interprets labor supply outcomes. The distinction between ranking and tool demand is useful: the ranking exercise shows that the video transmits a general message, while the tool-demand effect shows that women who learn they are worse off than expected become more interested in personalized planning.

Mechanism role: The table separates information acquisition from labor supply. The labor supply effects would be less credible if the treatment did not first change understanding or planning behavior. The table also supports heterogeneity by prior beliefs: the treatment is more behaviorally relevant when it corrects an optimistic prior.

Reading caution: A financial index can mask differences across components. The economic interpretation should focus on both the aggregate index and the specific columns: ranking effects capture understanding, while tool take-up captures follow-up demand and revealed interest.

Economic message: The intervention is not merely a motivational video; it changes what mothers know and what financial tools they want to use. That makes later changes in hours interpretable as information-driven reoptimization.

TABLE I
TREATMENT IMPACT ON FINANCIAL OUTCOMES

	Wave 1				Follow-up	
	Financial index (1)	Correct ranking (2)	Tools index (3)	Consultation (4)	Correct ranking (5)	Fin. long-term (6)
Panel A: Main estimates						
Treat	0.392***	0.313***	0.088*	0.020	0.226***	−0.020
	(0.046)	(0.022)	(0.050)	(0.023)	(0.029)	(0.020)
	[0.001]	[0.001]	[0.062]	[0.149]	[0.001]	[0.135]
Spillover	0.008	0.003	−0.004	0.013	0.034	−0.018
	(0.050)	(0.025)	(0.051)	(0.023)	(0.030)	(0.019)
	[1.000]	[1.000]	[1.000]	[1.000]	[1.000]	[1.000]
Panel B: Heterogeneity						
Underestimators						
Treat	0.569***	0.297***	0.313***	0.069	0.279***	0.038
	(0.093)	(0.052)	(0.103)	(0.048)	(0.063)	(0.046)
	[0.001]	[0.001]	[0.002]	[0.066]	[0.001]	[0.158]
Spillover	−0.012	−0.028	0.009	0.015	0.087	−0.005
	(0.104)	(0.057)	(0.107)	(0.048)	(0.064)	(0.042)
	[1.000]	[1.000]	[1.000]	[1.000]	[1.000]	[1.000]

TABLE I
CONTINUED

	Wave 1				Follow-up	
	Financial index (1)	Correct ranking (2)	Tools index (3)	Consultation (4)	Correct ranking (5)	Fin. long-term (6)
Overestimators						
Treat	0.344***	0.320***	0.022	0.007	0.205***	−0.037*
	(0.056)	(0.024)	(0.061)	(0.028)	(0.035)	(0.021)
	[0.001]	[0.001]	[0.362]	[0.362]	[0.001]	[0.070]
Spillover	0.040	0.013	0.022	0.014	0.013	−0.025
	(0.057)	(0.029)	(0.059)	(0.029)	(0.036)	(0.022)
	[1.000]	[1.000]	[1.000]	[1.000]	[1.000]	[1.000]
Adjusted R^2	0.08	0.12	0.07	0.03	0.04	0.00
No. observations	2,216	2,216	2,216	2,216	1,656	1,642
Pure control mean	−0.00	0.54	−0.00	0.29	0.54	0.13
Pure control mean (under)	−0.06	0.54	−0.08	0.27	0.52	0.12
Pure control mean (over)	0.03	0.54	0.04	0.30	0.55	0.14
p-value	.05	.69	.02	.28	.32	.14

Notes. This table shows treatment and spillover effects on financial outcomes. Column (1): financial index, which aggregates the correct ranking (column (2)) and the tools index (column (3)). Columns (2) and (5): indicator for whether respondents correctly rank total future salary and pension savings above child care costs using the part-time vignette (see Section III.A). Column (3): tools index, measures the willingness to sign up to receive different information materials and resources related to financial planning, including the incentivized sign-up for a financial consultation. Column (4): indicator for signing up for a financial consultation (incentivized). Column (6): indicator for mentioning long-term financial factors in an open-ended text question about the most important factors considered for employment-level plans in 10 years. Data: wave 1 (columns (1)–(4)) and follow-up (column (5)–(6)). Panels A and B are estimated in separate regressions. Panel A: estimates of equation (1). Panel B interacts the treatment indicators in equation (1) with group indicators for cost under- versus overestimation and adds the group indicator to the set of controls. The adjusted R^2 is reported for Panel B. The control mean is reported for pure control group overall, under-, and overestimators. The p-value is for a test of equality of coefficients between under- and overestimators in the treatment group. All specifications use strata fixed effects and post-double-selection LASSO to determine the set of controls (Belloni, Chernozhukov, and Wei 2016), listed in Online Appendix Table A.5.2. Standard errors clustered at the school level are in parentheses, and sharpened q-values (Anderson 2008) for each row are in square brackets. * $p < .10$, ** $p < .05$, *** $p < .01$.

Table I: Treatment impact on financial outcomes

5.3 Figure II: Nonparametric Heterogeneous Treatment Effects by Pension Estimate

Read:

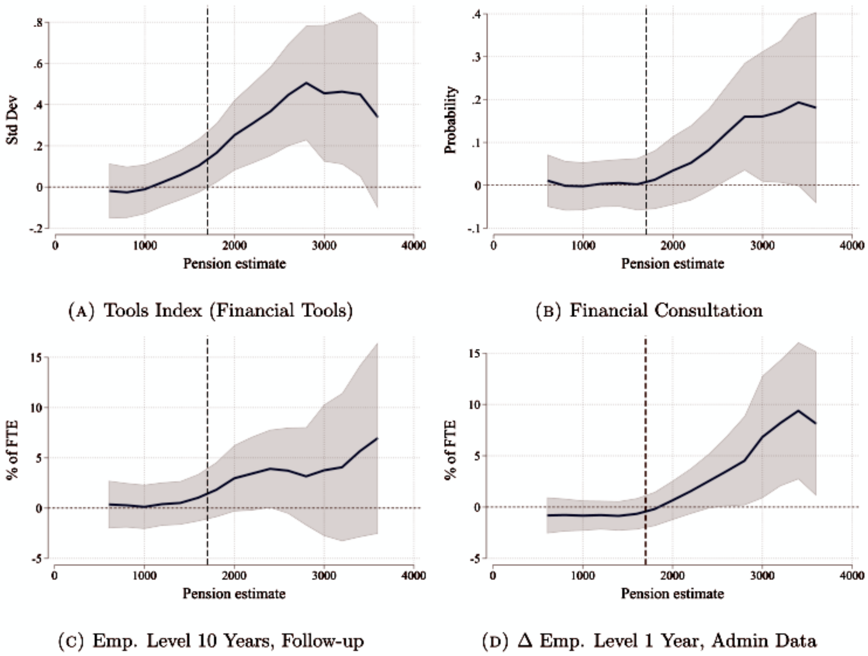
- Figure II plots local treatment effects against the respondent's baseline estimate of pension receipt under part-time work.
- The vertical reference line marks the projected value from the Future Calculator.
- Panels cover demand for tools, sign-up for consultation, 10-year employment plans, and administrative one-year employment changes.
- Effects rise primarily on the side of the distribution where women overestimate pension receipt, meaning where part-time work looks too financially benign at baseline.
- The administrative panel mirrors the planning and financial-tool panels: women with optimistic pension beliefs are the ones whose actual hours rise.

Interpretation: This figure is one of the strongest mechanism exhibits because it moves beyond a discrete underestimator/overestimator split. It shows that treatment effects vary continuously with the degree of baseline misperception. That pattern is difficult to reconcile with a simple "all treated women become more motivated" story.

Mechanism role: The shape of the curves supports belief updating. The treatment is most effective where the information is most corrective, especially for women whose baseline pension estimates are too high.

Reading caution: Nonparametric curves are descriptive and can be noisy at the tails of the pension-estimate distribution. The shaded confidence bands should be read together with the regression tables.

Economic message: The figure shows that the information intervention targets a real margin of misperception: the larger the optimistic pension error, the more the treatment shifts planning and behavior.



(A) Tools Index (Financial Tools)

(B) Financial Consultation

(C) Emp. Level 10 Years, Follow-up

(D) Δ Emp. Level 1 Year, Admin Data

Figure II: Nonparametric heterogeneous treatment effects by pension estimate

5.4 Table II: Treatment Impact on Short-Term Labor Supply Outcomes

Read:

- Table II studies planned employment for the next academic year, follow-up plans, and administrative employment changes one year after treatment.
- In the full sample, treated teachers report a modest increase in next-year employment plans.
- The effect is much larger for cost underestimators: their planned increase is about 4.95 percentage points, around 9% over the pure-control mean.
- The administrative outcome shows that treated cost underestimators increase actual employment by 3.87 percentage points one year later.
- The administrative effect is close to the planned increase, suggesting that plans were not cheap talk.
- Spillover estimates are positive for cost underestimators but smaller and less precisely estimated, consistent with partial information diffusion.

Interpretation: This table is the main behavioral evidence. It shows that the intervention affects not only financial awareness but actual labor supply. The cost-underestimator subgroup is the key group: they are exactly the women for whom the treatment reveals that part-time work is more costly than expected.

Mechanism role: The match between survey plans and administrative changes is important. It suggests that the treatment changed the decision, not merely the stated attitude. Because the administrative outcome is recorded one year later, it is less vulnerable to immediate experimenter-demand effects.

Reading caution: The administrative effect applies to Department of Education employment. The paper argues that this captures the main job for most teachers, but it cannot perfectly measure hours in outside employment.

Economic message: Correcting long-run cost beliefs changes a real labor supply margin. The result is sizable because a few percentage points of FTE translate into additional weekly teaching time and, if persistent, meaningful pension and lifetime income gains.

TABLE II

TREATMENT IMPACT ON SHORT-TERM LABOR SUPPLY OUTCOMES

	Δ*Employment level* 1 year		
	Wave 1 (1)	Follow-up (2)	Admin (3)
Panel A: Main estimates			
Treat	1.692**	0.141	−0.077
	(0.775)	(0.911)	(0.616)
	[0.098]	[1.000]	[1.000]
Spillover	1.275	0.522	0.061
	(0.800)	(0.894)	(0.582)
	[0.503]	[1.000]	[1.000]
Panel B: Heterogeneity			
Underestimators			
Treat	4.952***	3.536*	3.873***
	(1.694)	(2.000)	(1.321)
	[0.006]	[0.027]	[0.006]
Spillover	2.849*	4.238**	1.454
	(1.661)	(1.896)	(1.280)
	[0.096]	[0.085]	[0.150]
Overestimators			
Treat	0.967	−0.649	−0.999
	(0.919)	(1.042)	(0.748)
	[0.786]	[0.786]	[0.786]
Spillover	1.165	−0.451	0.065
	(0.984)	(1.084)	(0.726)
	[1.000]	[1.000]	[1.000]
Adjusted R^2	0.13	0.10	0.06
No. observations	2,302	1,687	2,152
Pure control mean (level)	54.75	55.37	53.30
Pure control mean	3.20	3.86	0.97
Pure control mean (under)	0.43	0.86	−1.42
Pure control mean (over)	3.69	4.41	1.56
p-value	.04	.07	.00

Notes. This table shows treatment and spillover effects on short-term labor supply. Column (1): change in next academic year's planned employment level in wave 1. Column (2): change in next academic year's planned employment level measured in the follow-up. Column (3): change in next academic year's employment level in the administrative data. All changes are relative to the employment level in administrative data at the time of the intervention (2022). Employment levels are measured as percent of a full-time equivalent. Panel A and B are estimated in separate regressions. Panel A: estimates of equation (1). Panel B interacts the treatment indicators in equation (1) with group indicators for cost under- versus overestimation and adds the group indicator to the set of controls. The adjusted R^2 is reported for Panel B. The control mean in levels is reported for the pure control group. Control mean (Δ*Employment level*) is reported for the pure control group overall, under-, and overestimators. The p-value is for a test of equality of coefficients between under- and overestimators in the treatment group. All specifications use strata fixed effects and post-double-selection LASSO to determine the set of controls (Belloni, Chernozhukov, and Wei 2016), listed in Online Appendix Table A.5.2. Standard errors are clustered at the school level in parentheses, and sharpened q-values (Anderson 2008) for each row are reported in square brackets. * $p < .10$, ** $p < .05$, *** $p < .01$.

Table II: Treatment impact on short-term labor supply outcomes

5.5 Table III: Treatment Impact on Planned Long-Term Labor Supply Outcomes

Read:

- Table III studies medium- and long-run employment plans at three, five, and ten years.
- The overall sample shows some positive movement, but many full-sample coefficients are noisy.
- Cost underestimators plan meaningful upward adjustments over multiple horizons.
- For cost underestimators, the paper reports around four percentage-point increases at five- and ten-year horizons, corresponding to roughly 6–7% over the pure-control mean.

- Overestimators or women with accurate/low estimates do not show a symmetric sustained decrease in planned labor supply.

Interpretation: Table III addresses persistence. A one-year administrative increase might be interpreted as a temporary adjustment, but the stated medium- and long-run plans suggest that cost underestimators revise their intended employment path upward, not merely the next year's contract.

Mechanism role: The table supports an intertemporal reoptimization channel. The treatment causes women who underestimated costs to reconsider future labor supply in a way that is consistent with the lifetime-income and pension logic of the intervention.

Reading caution: Long-run plans are still survey outcomes. They are informative about intended persistence, but they are not the same as observed decade-long employment histories.

Economic message: If the planned increase persists, the treatment could have effects far beyond the first year by raising lifetime earnings, pension accumulation, and financial independence.

TABLE III

TREATMENT IMPACT ON PLANNED LONG-TERM LABOR SUPPLY OUTCOMES

	Incentive compatible, follow-up			Follow-up	Wave 1	Follow-up
Employment level	3 years (1)	5 years (2)	10 years (3)	10 years (4)	10 years (5)	index (6)
Panel A: Main estimates						
Treat	0.564	1.006	0.663	1.063	3.129***	0.030
	(0.681)	(0.792)	(0.989)	(0.887)	(0.695)	(0.050)
	[0.840]	[0.629]	[0.840]	[0.629]	[0.001]	[0.840]
Spillover	0.390	−0.068	0.122	0.131	0.066	0.009
	(0.692)	(0.786)	(0.993)	(0.875)	(0.693)	(0.051)
	[1.000]	[1.000]	[1.000]	[1.000]	[1.000]	[1.000]
Panel B: Heterogeneity						
Underestimators						
Treat	3.136*	4.351**	4.593*	4.335**	4.833***	0.274**
	(1.765)	(1.903)	(2.381)	(1.885)	(1.530)	(0.117)
	[0.036]	[0.030]	[0.036]	[0.030]	[0.011]	[0.030]
Spillover	2.087	0.428	−1.616	−1.731	0.619	0.003
	(1.767)	(1.819)	(2.241)	(1.885)	(1.614)	(0.115)
	[1.000]	[1.000]	[1.000]	[1.000]	[1.000]	[1.000]

TABLE III

CONTINUED

	Incentive compatible, follow-up			Follow-up	Wave 1	Follow-up
Employment level	3 years (1)	5 years (2)	10 years (3)	10 years (4)	10 years (5)	index (6)
Overestimators						
Treat	−0.019	−0.058	−0.501	0.478	2.448***	−0.035
	(0.806)	(0.895)	(1.059)	(0.955)	(0.843)	(0.055)
	[1.000]	[1.000]	[1.000]	[1.000]	[0.024]	[1.000]
Spillover	0.060	−0.255	0.441	0.969	−0.186	0.015
	(0.802)	(0.902)	(1.091)	(1.000)	(0.881)	(0.057)
	[1.000]	[1.000]	[1.000]	[1.000]	[1.000]	[1.000]
Adjusted R^2	0.50	0.39	0.15	0.18	0.20	0.33
No. observations	1,652	1,641	1,636	1,684	2,295	1,626
Pure control mean	57.24	61.68	68.28	69.12	70.11	−0.00
Pure control mean (under)	55.04	60.17	67.33	68.60	69.01	−0.12
Pure control mean (over)	57.89	62.23	68.76	69.19	70.65	0.04
p-value	.12	.04	.04	.06	.19	.02

Notes. This table shows treatment and spillover effects on planned long-term labor supply. Columns (1)–(3): incentive-compatible elicitation of planned employment level in 3, 5, and 10 years for Department of Education measured in the follow-up. Column (4): planned employment level in 10 years measured in the follow-up (any employer). Column (5): planned employment level in 10 years measured in the wave 1 survey. Column (6): index across all long-term employment-level measures in the follow-up survey. For the incentive-compatible elicitation, we informed participants that their answers would be used to generate a forecast of the teacher workforce for the Department of Education. Employment levels are measured as percent of a full-time equivalent. Panels A and B are estimated in separate regressions. Panel A: estimates of equation (1). Panel B interacts the treatment indicators in equation (1) with group indicators for cost under- versus overestimation and adds the group indicator to the set of controls. The adjusted R^2 is reported for Panel B. The control mean is reported for the pure control group overall, and for under- versus overestimators. The p-value is for a test of equality of coefficients between under- and overestimators in the treatment group. All specifications use strata fixed effects and post-double-selection LASSO to determine the set of controls (Belloni, Chernozhukov, and Wei 2016), listed in Online Appendix Table A.5.2. Standard errors clustered at the school level are in parentheses, and sharpened q-values (Anderson 2008) for each row are reported in square brackets. * $p < .10$, ** $p < .05$, *** $p < .01$.

Table III: Treatment impact on planned long-term labor supply outcomes

5.6 Figure III: Change in Labor Supply by Cost Underestimation, Administrative Data

Read:

- Figure III plots the raw distribution of changes in employment level from 2022 to 2023.
- The left panel shows cost overestimators; the right panel shows cost underestimators.
- Among overestimators, the treated and pure-control distributions look very similar.
- Among underestimators, the treated distribution shifts toward positive employment changes, with visible mass around a 10 percentage-point increase.
- A 10 percentage-point increase corresponds roughly to an extra half-day of work.

Interpretation: The figure visually verifies the administrative result in Table II. The effect is not simply a regression artifact: the raw distribution moves in the expected subgroup and does not move in the nonresponsive subgroup.

Mechanism role: The subgroup contrast is essential. Cost underestimators receive bad news relative to their priors and increase hours; overestimators learn that part-time work may be less costly than they feared but do not reduce hours. This asymmetric response is consistent with information being more actionable when it corrects optimistic mistakes.

Reading caution: The figure excludes extreme changes outside the plotted range and is descriptive. The regression table remains necessary for controls, clustering, and formal inference.

Economic message: The treatment changes actual employment in the data system that determines salaries and pension contributions. This is the paper's central behavioral result.

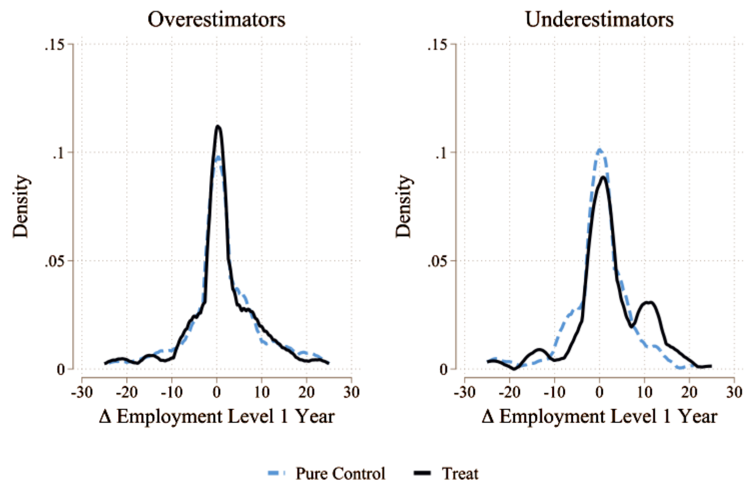


FIGURE III

Figure III: Change in labor supply by cost underestimation, administrative data

5.7 Table IV: Mechanisms: Reactions to Treatment

Read:

- Table IV studies emotional responses, information sharing, planning actions, household effects, and related outcomes.
- The treatment initially produces negative emotions, especially among cost underestimators, consistent with the information being an unpleasant correction.
- Treated women are more likely to discuss the material with partners, colleagues, or social contacts.
- The treatment increases action-taking, such as engaging with financial planning or considering employment changes.
- The paper finds no strong evidence of effects on partners' employment or fertility plans.
- Cost underestimators tend to be less interested in financial topics and more gender-conservative, which helps explain why they may have remained uninformed before the intervention.

Interpretation: Table IV helps distinguish belief updating from pure salience. A salience story would mainly say that the treatment reminds women to think about money. The mechanism evidence instead suggests that some women learn inconvenient information, discuss it, and then act on it. The emotional response is meaningful because the information reveals that prior decisions may have carried larger costs than expected.

Mechanism role: The table connects beliefs to social transmission and action. It explains why spillovers are plausible: treated teachers discuss the content with others. It also explains why the effect is concentrated among underestimators: they had lower financial interest and more traditional attitudes at baseline, so the information was more novel.

Reading caution: Mechanism outcomes are mostly survey-based, so they are more vulnerable to reporting concerns than administrative hours. Their role is to interpret the behavioral response, not to replace the administrative evidence.

Economic message: The treatment works by making hidden future costs concrete enough to trigger reflection, conversation, and reoptimization. This is why a low-cost information intervention can have a real labor supply effect.

TABLE IV

MECHANISMS: REACTIONS TO TREATMENT

	Emotions index	Stress index	Talk to			Take
	Wave 1 (1)	Follow-up (2)	Anybody (3)	Partner (4)	Colleague (5)	Action (6)
Panel A: Main estimates						
Treat	−0.412***	−0.124**	0.196***	0.177***	0.116***	0.124***
	(0.051)	(0.060)	(0.029)	(0.027)	(0.021)	(0.022)
	[0.001]	[0.007]	[0.001]	[0.001]	[0.001]	[0.001]
Spillover	−0.039	0.050	−0.053*	−0.026	0.011	0.011
	(0.049)	(0.057)	(0.029)	(0.026)	(0.017)	(0.016)
	[0.800]	[0.800]	[0.636]	[0.800]	[0.800]	[0.800]
Panel B: Heterogeneity						
Underestimators						
Treat	−0.680***	−0.035	0.141**	0.115*	0.101**	0.133***
	(0.113)	(0.130)	(0.067)	(0.066)	(0.051)	(0.051)
	[0.001]	[0.152]	[0.049]	[0.073]	[0.049]	[0.025]
Spillover	−0.081	0.053	−0.160**	−0.115*	−0.053	−0.014
	(0.112)	(0.127)	(0.067)	(0.065)	(0.041)	(0.040)
	[0.637]	[0.637]	[0.124]	[0.237]	[0.351]	[0.637]
Overestimators						
Treat	−0.337***	−0.117*	0.211***	0.187***	0.120***	0.134***
	(0.062)	(0.067)	(0.033)	(0.031)	(0.023)	(0.024)
	[0.001]	[0.014]	[0.001]	[0.001]	[0.001]	[0.001]
Spillover	−0.028	0.055	−0.015	−0.001	0.030	0.022
	(0.060)	(0.067)	(0.033)	(0.030)	(0.020)	(0.019)
	[1.000]	[1.000]	[1.000]	[1.000]	[1.000]	[1.000]
Adjusted R^2	0.08	0.08	0.07	0.07	0.04	0.03

TABLE IV

CONTINUED

	Emotions index	Stress index	Talk to			Take
	Wave 1 (1)	Follow-up (2)	Anybody (3)	Partner (4)	Colleague (5)	Action (6)
No. observations	2,281	1,669	1,659	1,645	1,638	1,659
Pure control mean	−0.00	−0.00	0.38	0.30	0.08	0.08
Pure control mean (under)	0.16	−0.01	0.47	0.38	0.14	0.10
Pure control mean (over)	−0.05	−0.01	0.36	0.28	0.07	0.08
p-value	.01	.56	.35	.33	.74	.99

Notes. This table shows treatment and spillover effects on emotions and actions in response to the treatment. Column (1): emotions index measured in wave 1, with positive values indicating more positive emotions. Index across indicators for whether the respondent feels angry, anxious, hopeful, discouraged, happy, or motivated about their future. Column (2): stress index using a reduced version of the Perceived Stress Scale, with positive values indicating higher levels of stress. Columns (3)–(5): indicator for talking about the content of the video to anybody, their partner/family, or colleagues. Column (6): indicator for planning to take any action in response to the video. Column (1) based on data from wave 1, columns (2)–(6) based on data from the follow-up survey. Panels A and B estimated in separate regressions. Panel A: estimates of equation (1). Panel B interacts the treatment indicators in equation (1) with group indicators for cost under- versus overestimation and adds the group indicator to the set of controls. The adjusted R^2 is reported for Panel B. The control mean is reported for the pure control group overall, and for under- versus overestimators. The p-value is for a test of equality of coefficients between under- and overestimators in the treatment group. All specifications use strata fixed effects and post-double-selection LASSO to determine the set of controls (Belloni, Chernozhukov, and Wei 2016), listed in Online Appendix Table A.5.2. Standard errors clustered at the school level are in parentheses, and sharpened q-values (Anderson 2008) for each row are reported in square brackets. * $p < .10$, ** $p < .05$, *** $p < .01$.

Table IV: Mechanisms: reactions to treatment

6 Short summary

- Long-run financial consequences are rarely top of mind when mothers choose labor supply after childbirth.
- A substantial share of mothers holds overly optimistic beliefs about pension or wage-growth consequences of part-time work.
- The Swiss teacher setting allows accurate projections because salary schedules are deterministic and hours can be changed annually.
- The experiment uses two-stage randomization to estimate both direct treatment effects and social spillovers.
- The treatment improves financial understanding and raises demand for planning tools.
- Effects are concentrated among cost underestimators, whose priors make the treatment information most corrective.
- Cost underestimators revise short-run employment plans upward and increase actual employment one year later by 3.87 percentage points.
- Long-run plans suggest that cost underestimators expect to sustain higher employment paths.
- Spillover evidence is suggestive but noisy, consistent with partial social transmission of the treatment message.
- Mechanism evidence shows negative initial emotions, more discussion, and more planning actions.
- The policy message is that families may need accurate, personalized projections to fully evaluate the trade-offs created by childcare policies, part-time work, and pension systems.

7 Conclusion

7.1 Core conclusion

The paper shows that information frictions can shape maternal labor supply. When mothers underestimate the long-run financial costs of part-time work, objective personalized information can change plans and actual hours. The key empirical contribution is that the paper links survey-based belief correction to employer administrative outcomes one year later.

7.2 Economic mechanism

The mechanism is an intertemporal information friction. Immediate childcare needs, household logistics, family preferences, and short-run finances are salient, while lifetime earnings and pension consequences are delayed, complex, and hard to calculate. The intervention makes these delayed costs visible and personally relevant. The response is strongest among cost underestimators because the treatment changes their perceived trade-off most sharply.

7.3 Evidence chain

The paper's evidence chain has five steps. First, descriptive survey evidence shows that long-run financial considerations are rarely top of mind. Second, vignette-based elicitation shows that many women hold overly optimistic beliefs about pension and wage-growth costs of part-time work. Third, randomized treatment improves understanding and financial-tool demand. Fourth, cost underestimators revise short- and long-run labor supply plans upward. Fifth, administrative data show that the same group increases contracted employment one year later.

7.4 Why the result matters

The result matters because many policies aim to increase maternal labor supply by reducing external constraints, such as childcare costs or work-schedule rigidities. This paper shows that those policies may be less effective if families do not perceive the long-run financial benefits of increasing hours. Information does not replace childcare subsidies or workplace flexibility, but it can help families solve the correct optimization problem when those constraints are relaxed.

7.5 What not to overinterpret

The paper does not show that all mothers should work more, nor that information alone can close gender gaps. Some mothers may fully understand the long-run costs and still choose lower hours for family, health, preference, or household-bargaining reasons. The teacher setting is also favorable: salaries are predictable, hours are adjustable, and pension implications can be projected accurately. In occupations with uncertain promotions or rigid schedules, information may be less immediately actionable.

7.6 Policy implication

Employers, pension funds, schools, and governments could provide default personalized projections of how hours choices affect lifetime earnings, pension receipt, and financial resilience. The treatment is low cost and scalable, especially when projections can be automated. The paper also suggests that delivering such information through social or workplace networks may create multiplier effects, although these spillovers need further evidence.

7.7 Broader implication

The paper contributes to behavioral public economics by showing that inattention and mistaken beliefs about delayed financial consequences can matter even in high-stakes labor supply decisions. It also contributes to the gender-gap literature by identifying an information channel behind persistent part-time work. The findings imply that remaining gender gaps are not only the result of constraints and preferences; they may also reflect incomplete or non-salient information about the future.

7.8 Final takeaway

The result is not that every mother should increase hours. The result is that hours decisions should be made with the full long-run financial trade-off visible. When that trade-off becomes clear, some mothers—especially those who had underestimated the cost of part-time work—choose to work more.